

NOAH NDAMBUKI

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BILINGUAL: ENGLISH, FRENCH. CANADIAN CITIZEN.

EDUCATION

University of British Columbia

Bachelor of Applied Science in Computer Engineering (Dean's List: 84%)

Expected Graduation: May 2027

Vancouver, BC

- **Relevant Coursework:** Applied Machine Learning, Software Construction I & II, Operating Systems, Algorithms and Data Structures, Computing Systems I & II, Computer Communications, CPEN Design Studio.

TECHNICAL EXPERIENCE

Grid Insights | 🌐 posytt.com

May 2025 – Present

Lead Software Developer (16-month Co-op, Part-Time from September 2026)

Vancouver, BC

- Lead developer for the **Azure** platform behind a browser-based power grid product: a **Python Azure Functions** API of **54 endpoints** over **Azure SQL** via **managed identity**, with **CI/CD** gating every deploy on **2,100+ tests**.
- Cut peak browser heap **28%** and network-diff latency **28%** on a **100k+** bus case by indexing the diff baseline instead of holding a **196 MB** snapshot.
- Developed the in-app **AI assistant** in **React/TypeScript**: a **16-step agentic loop** over **5 LLMs** and **62 tools** that edit the grid model and drive a **Rust/WebAssembly** solver client-side.
- Shipped **Stripe** monetization end to end (checkout, **signed webhooks**, subscriptions, refunds) over an **idempotent credit ledger** metering **LLM** spend across **7 endpoints**, with plans granting and revoking **SQL** role entitlements.

JuaIT | 🌐 juait.com

March 2026 – Present

Co-Founder / Software Lead

Vancouver, BC

- Engineered a bilingual English/Swahili **WhatsApp** assistant in **Python/FastAPI**: an **LLM parser** feeds a **deterministic state machine**, and an engine sizes and ranks solar irrigation systems from satellite irradiance data; deployed to **Azure Container Apps** with **77 automated tests**.
- Founded the venture and co-authored its **24-page concept paper**; selected for **UBC Venture Founder**, taking the product from concept to a **working MVP** in **six months** as sole developer, now preparing a farmer pilot in Kenya.

UBC Open Robotics | 🌐 openrobotics.ca

September 2025 – Present

Software Developer

Vancouver, BC

- Architected **concurrent** real-time firmware for a guitar tuner in **C++** on **ESP32**: a **4-stage producer-consumer pipeline** of **FreeRTOS** tasks pinned across **2 cores**, passing **128 ms** frames over **3 single-slot queues**.
- Built the **pitch detection** and **PID control** modules: **autocorrelation** over a **60 to 400 Hz** band at **8 kHz** with integral clamping and an output deadband.

PROJECTS

The 30-Minute Chess Machine | 🌐 canyoubeat chess.com | Python, PyTorch, ONNX, TypeScript, React, Azure, CI/CD

- Trained a **1.7M-parameter policy/value network** in **PyTorch** by **knowledge distillation** from Stockfish 18 over **7.8M labelled positions** on **AWS EC2**; served client-side via **ONNX Runtime Web** and **WebAssembly SIMD** at **46.7 ms p95** inference on a mid-range Android phone (**\$0 serving cost**).
- Shipped a **React/TypeScript** site on **Azure Static Web Apps** offering **five playable model checkpoints** from a **510-checkpoint** run, gated by **1,000+ tests** including **move-for-move TypeScript-Python parity**.
- Cut the in-browser opponent's **per-move decision time** from **640 ms** to **23 ms (28x)** by reading the best reply from a **closed-form material bound** and parsing **SAN**, verified identical on **5,378 positions**.

OddsSea | 🌐 oddssea.xyz | TypeScript, React, AWS, PostgreSQL, Lambda, CDK, CI/CD

- Built an **AWS** backend on **Lambda** and **Aurora Serverless v2 PostgreSQL**, defined as **three CDK IaC** stacks and released on every merge through **GitHub Actions OIDC** with **zero stored credentials**.
- Implemented **seven real-time wagering games** settling into an **append-only PostgreSQL ledger**, deriving every outcome **deterministically** from **HMAC-seeded round secrets** for **independent auditability**.

TECHNICAL SKILLS

Frameworks & Technologies: React, FastAPI, PyTorch, ONNX, PostgreSQL, Stripe, Git, pytest, Vitest, REST APIs, WASM

Cloud & DevOps: AWS, Azure, Docker, IaC (Bicep, AWS CDK), GitHub Actions CI/CD

Languages: Python, TypeScript, JavaScript, C, C++, Java, SQL, HTML, CSS